CIE AS PHYSICS (9702) • QUESTION PAPER

7.4 Electromagnetic Spectrum

Syllabus Section: 7. Waves **Paper Type:** Structured Questions (Paper 2) **Total:** 7 Questions (63 Marks)

Question 1 Easy [8 marks]

Part (a) [2 marks]

EM waves consist of oscillations in two perpendicular fields.

State the names of the fields.

Part (b) [3 marks]

Table 1.1 lists the names of the waves in the EM spectrum in order of increasing frequency.

Table 1.1

EM wave
Radio
Infrared
Ultraviolet
Gamma

Complete Table 1.1 by adding in the missing information.

Part (c) [2 marks]

State two uses for infrared radiation.

Part (d) [1 mark]

Give a value for the speed of gamma radiation travelling through a vacuum.

Question 2 Easy [12 marks]

Part (a) [3 marks]

Give the name of electromagnetic wave with the

(i) Lowest energy.

[1]

(ii) Shortest wavelength.

[1]

(iii) Lowest frequency.

[1]

Part (b) [2 marks]

State the range of wavelengths for infrared radiation.

Part (c) [4 marks]

Determine the range of frequencies for infrared radiation.

Part (d) **[3 marks]**

The visible region of the electromagnetic spectrum has frequencies ranging from 4.3×10^{14} Hz to 7.5×10^{14} Hz.

Show that the wavelength of visible light is 400 nm to 700 nm.

Question 3 Medium [4 marks]

Part (a) [1 mark]

List the electromagnetic waves in order of increasing wavelength.

.....

.....

Part (b) [2 marks]

State the two types of electromagnetic radiation that can knock electrons out of orbit.

.....

.....

.....

.....

Part (c) [1 mark]

Complete the following sentence about electromagnetic radiation by choosing the correct word.

wavelength	**frequency**	**amplitude^2**

The energy of an electromagnetic wave is directly proportional to .....

.....

.....

Question 4 Medium [12 marks]

Part (a) [3 marks]

Complete the table by stating the correct electromagnetic wave for each range of wavelengths.

EM wave	Range of wavelengths / m
	10^{-1} - 10^{6}
	4×10^{-7} - 7×10^{-7}
	10^{-16} - 10^{-10}

Part (b) [6 marks]

Determine the range of frequencies for the three waves given in part (a).

Part (c) [3 marks]

Suggest a reasonable estimate for the wavelength of green light. Explain your answer.

Question 5 Medium [9 marks]

Part (a) [1 mark]

State the three types of EM radiation that are used for communications.

Part (b) [4 marks]

State the properties common to all electromagnetic radiation.

Part (c) [4 marks]

Dogs can see wavelengths in the range of 430 - 475 nm which they perceive as blue, and 500 - 620 nm which they perceive as yellow.

Determine the range of frequencies which dog's eyes can detect.

Question 6 Hard [7 marks]

Part (a) [2 marks]

Justify the approximation of the speed of light in air as 3.00×10^{8} m s^{-1}.

Part (b) [2 marks]

Frequency and wavelength are fundamental properties of all waves. The relationship between them depends on the context of interaction and on the type of wave in question.

Sketch a graph showing the relationship between the frequency and wavelength of an electromagnetic wave.

Part (c) [3 marks]

Electromagnetic waves carry energy through the vacuum of space as progressive transverse waves.

Describe the motion of an electromagnetic wave with reference to the appropriate fields.

Question 7 Hard [11 marks]

Part (a) [2 marks]

Microwaves can be used in communications between satellites.

Calculate the frequency range for microwaves.

Part (b) [3 marks]

A satellite passing Neptune communicates with Earth using a microwave transmitter with an output power of 24.0 W and wavelength 78 900 μm.

The satellite's controller is located on Earth, at a distance of 4.40×10^{12} m when the signal is transmitted.

For this communication

(i) Calculate the time taken for the signal to be detected by the controller.

[1]

(ii) Calculate the energy of a microwave photon.

[2]

Part (c) [6 marks]

The controller dish aerial has an effective area of 258 m^2.

For the communication from part (b)

(i) Determine the power received by the controller dish aerial. You may assume that the power transmitted by the satellite radiates uniformly in all directions.

[2]

(ii) The actual power received by the controller dish aerial is 1.4×10^{-15} W. Suggest why this is different to the calculated power received.

[2]

(iii) Calculate the rate at which microwave photons arrive at the controller dish aerial.

[2]